

Consumer
Theory
[Lectures
1-5]

Consumer's Economic Circumstances

Lecture 1 (Nechyba's Chapter 2)

Key Assumptions in Microeconomics

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- In modelling micro-behaviour, we will assume that economic agents (individuals, firms,...) are **rational** and in pursuit of their perceived **self-interest**.
 - ▣ **Rational:** Economic agents seek to do the best they can given the circumstances (i.e. the constraint) they face.
 - ▣ **Self-interest:** Economic agents make decisions that increase their own welfare (they try to achieve their own “goals”).
- “Rationality” and “Self-Interest” are typically assumed in any *standard* microeconomic model.

Consumer Theory

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- The model we introduce today is a model that describes how an individual makes consumption choices (**consumer theory**).
- ▣ The framework we present is nevertheless more general and can be used to model how individuals make other economic choices, such as how many hours an individual will work and how much of her current income the individual will save to spend in consumption in the future.
- ▣ Consumer theory will keep us busy for 5 lectures.

Consumer Theory

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- How do economists model the way in which we (as consumers or individuals) make a choice?
- The underlying view is based on the two key assumptions of *rationality* and *self-interest*:
 - We choose what is *best for us* given what is *possible* for us.
 - What is **best** for us depends on our *tastes* (*lecture 2-3*)
 - What is **possible** depends on the **economic circumstances** that we face and that limit our choices.
(*today*)

Consumer Theory

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- Final choice can be seen as the result of a **two-step process**:
 - **First step**: Identify the **Choice Set**, i.e. the set of options that are available/feasible to you.
 - **Second step**: Pick the option in Choice Set that is the best for you (given your tastes).
 - This second step is formally known as **Constrained Utility Maximization** (lecture 4-5)
- Today we focus on the first step and learn how we can model the choice set of a consumer.
 - To do so we first introduce some assumptions that make our life easier.

A Consumer's Economic circumstances:

Assumptions

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- When you choose how much of a good to purchase, what limits your choice?
 - Income
 - Price
 - Law
 - Time
 - Etc...

- **Assumption 1:** We assume that the **economic circumstances** faced by a consumer are entirely described by the **income** of the consumer and the **prices** of the goods.

A Consumer's Economic circumstances:

Assumptions (cont'd)

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- **Assumption 2:** Consumers are **price takers**: they cannot influence the prices of the goods.

- **Assumption 3:** The income of a consumer is **exogenous**:
 - at the moment of making the choice, the consumer has an income whose amount has been previously determined and cannot be modified (the consumer takes his income as **given** and the only choice is how to spend it).

Budget Constraint and Choice Set

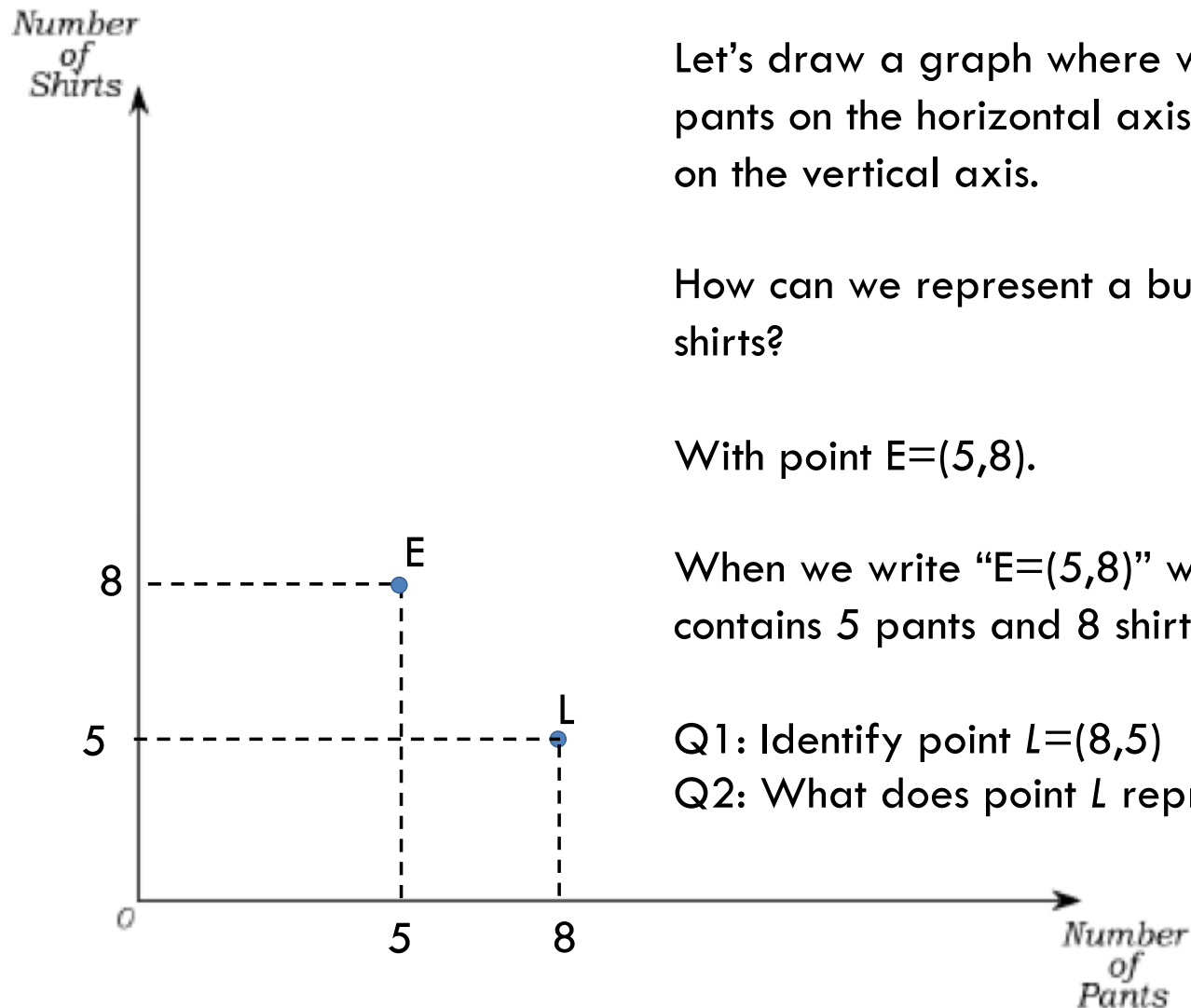
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- Suppose I have \$200 to spend on pants and shirts.
- The price of pants is \$20 and the price of shirts is \$10.
- Given those restrictions (i.e. the economic circumstances I face), I can afford only certain bundles or combinations of pants and shirts.
- The set of all bundles of shirts and pants that I can afford to buy is called my **Choice Set**.
- How do I *formally* represent a consumer's choice set?
 - ▣ Graphically
 - ▣ Mathematically

Budget Constraint and Choice Set

[Graph]

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Let's draw a graph where we measure the number of pants on the horizontal axis and the number of shirts on the vertical axis.

How can we represent a bundle of 5 pants and 8 shirts?

With point $E=(5,8)$.

When we write " $E=(5,8)$ " we mean "bundle E which contains 5 pants and 8 shirts".

Q1: Identify point $L=(8,5)$

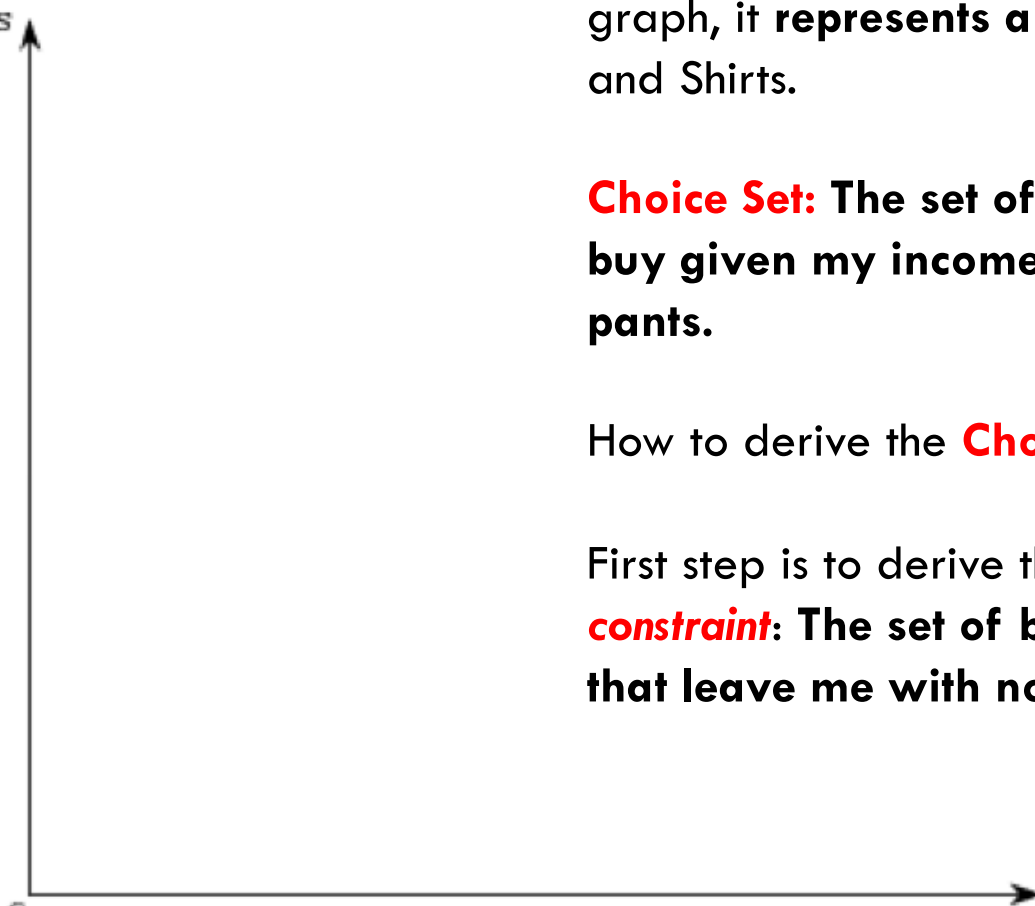
Q2: What does point L represent?

Budget Constraint and Choice Set

[Graph]

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Number
of
Shirts



In general, if you pick **a point** on this two-dimension graph, it **represents a bundle** or combination of Pants and Shirts.

Choice Set: The set of all bundles that I can afford to buy given my income and the prices of shirts and pants.

How to derive the **Choice set**?

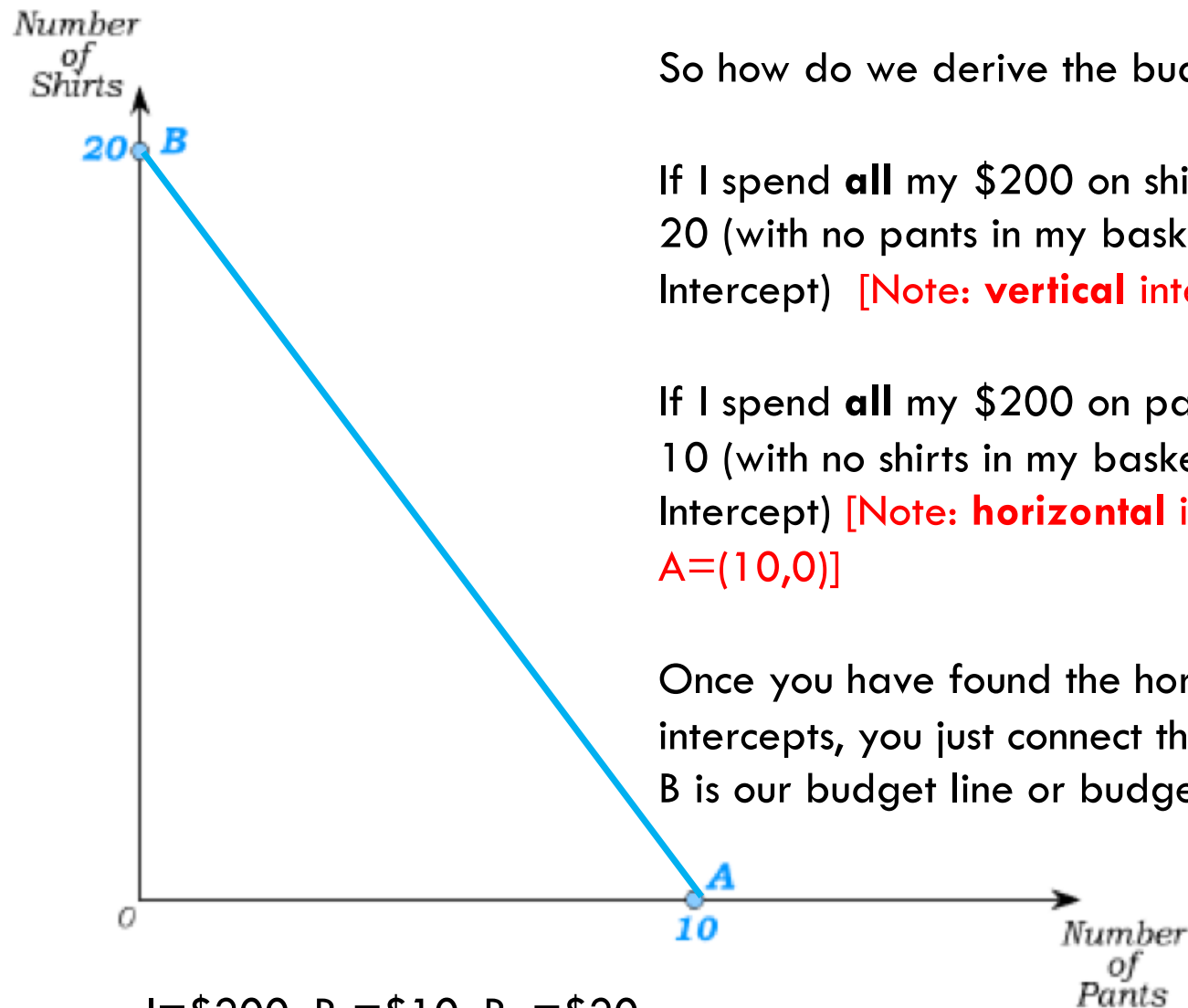
First step is to derive the **Budget line** or **budget constraint**: The set of bundles that are affordable and that leave me with no unspent income.

Number
of
Pants

Budget Constraint and Choice Set

[Graph]

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So how do we derive the budget constraint?

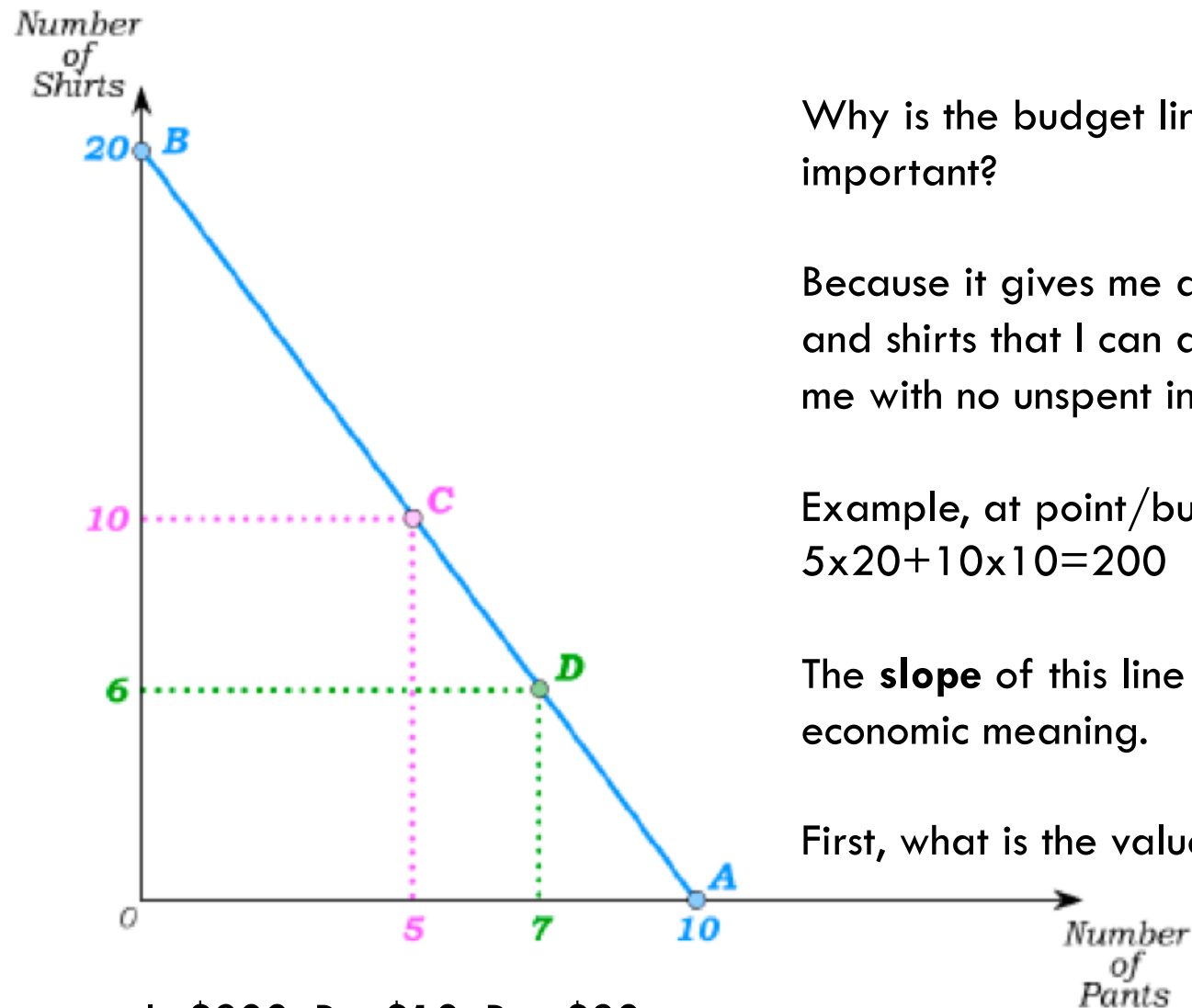
If I spend **all** my \$200 on shirts, I can buy as many as 20 (with no pants in my basket) – point **B** (Vertical Intercept) [Note: **vertical** intercept is bundle $B=(0,20)$]

If I spend **all** my \$200 on pants, I can buy as many as 10 (with no shirts in my basket) – point **A** (Horizontal Intercept) [Note: **horizontal** intercept is bundle $A=(10,0)$]

Once you have found the horizontal and vertical intercepts, you just connect them: the line connecting A to B is our budget line or budget constraint.

Budget Constraint and Choice Set [Graph]

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$I = \$200$, $P_s = \$10$, $P_p = \$20$

Why is the budget line or budget constraint important?

Because it gives me all the bundles of pants and shirts that I can afford to buy and leave me with no unspent income.

Example, at point/bundle $C = (5, 10)$, I spend:
 $5 \times 20 + 10 \times 10 = 200$

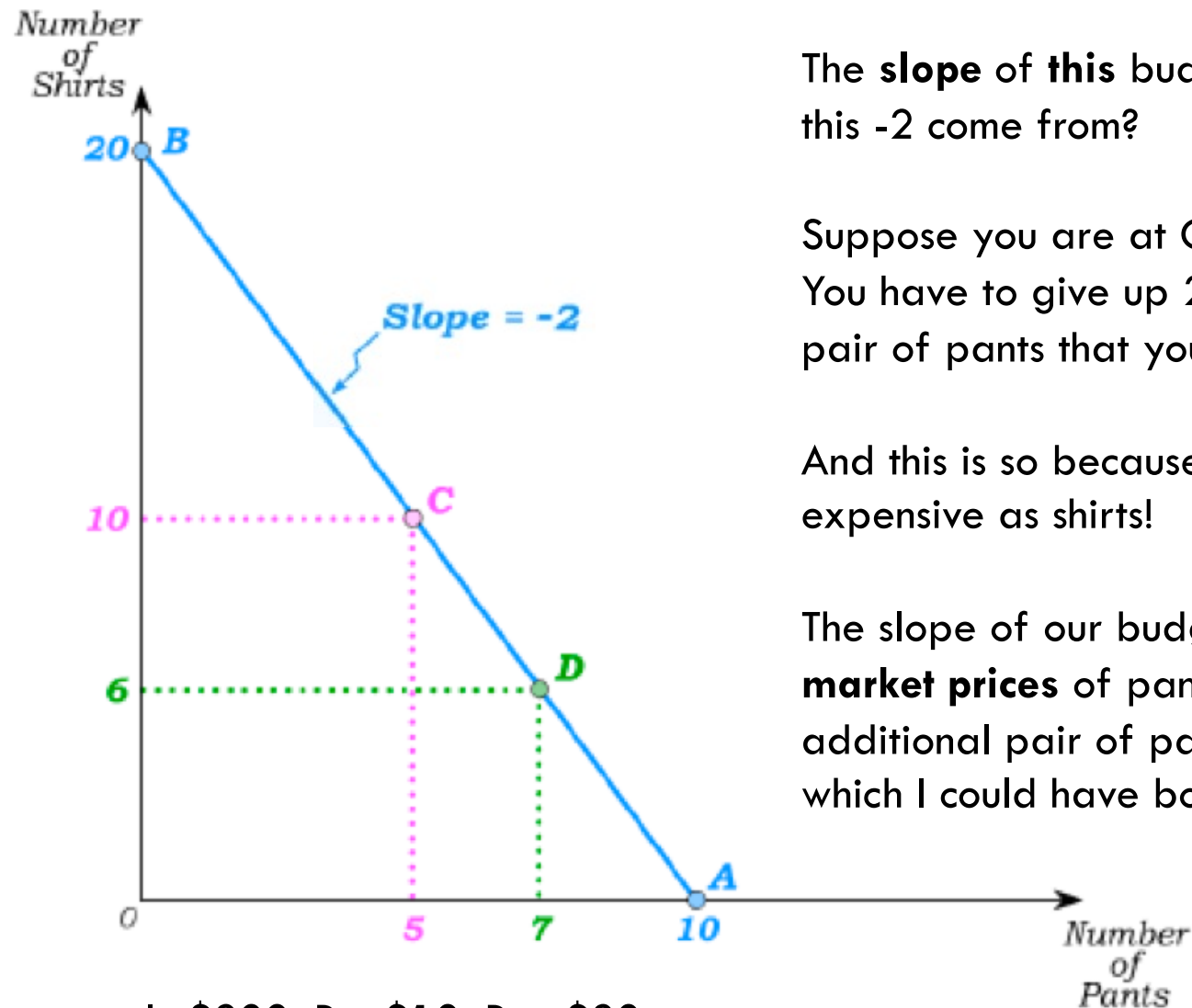
The **slope** of this line has an important economic meaning.

First, what is the value of the slope in **this** case?

Budget Constraint and Choice Set

[Graph]

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$I = \$200$, $P_s = \$10$, $P_p = \$20$

The **slope** of **this** budget line is -2 . Where does this -2 come from?

Suppose you are at C and want to move to D. You have to give up 2 shirts for every 1 additional pair of pants that you want to buy.

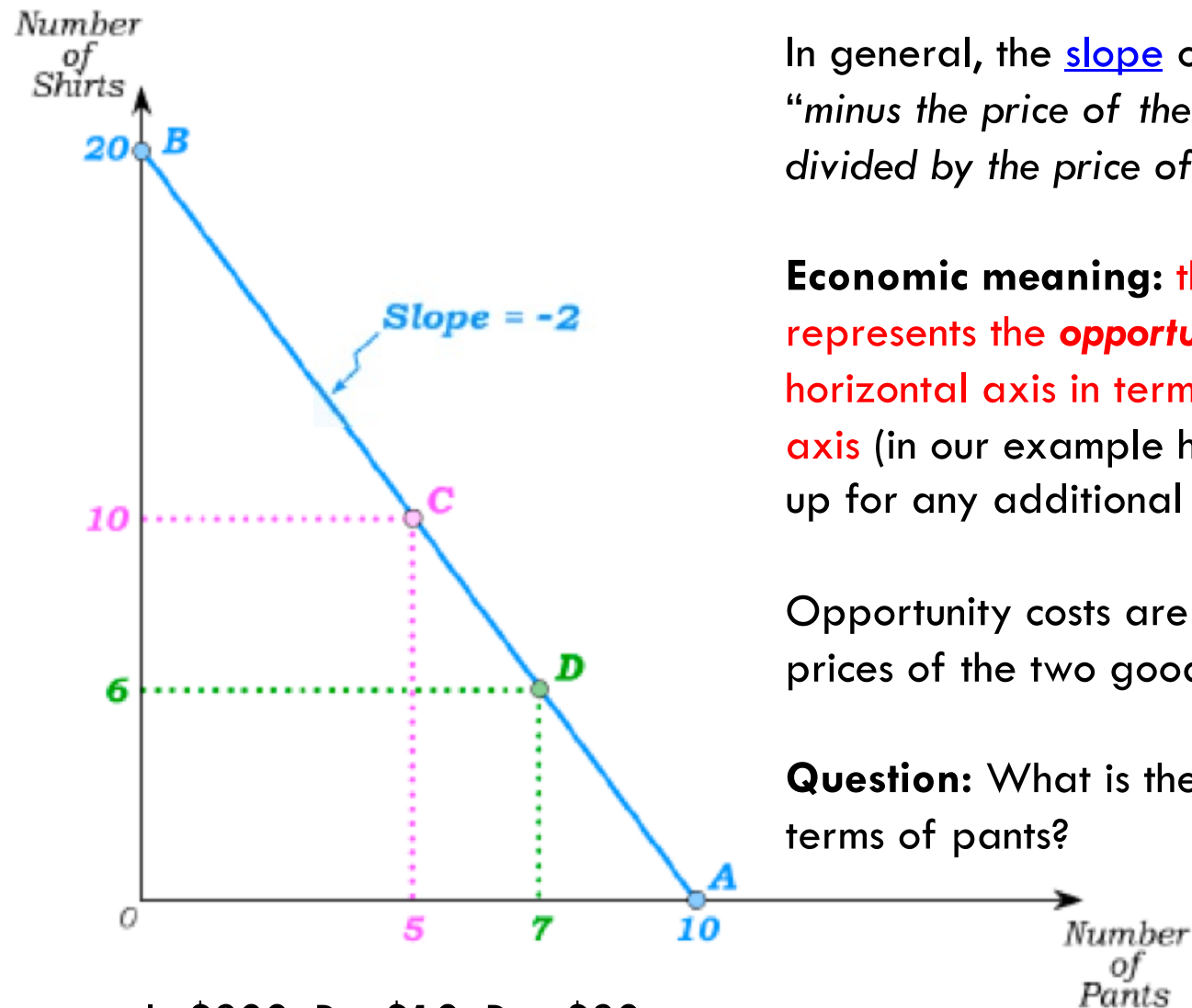
And this is so because pants are twice as expensive as shirts!

The slope of our budget line is determined by the **market prices** of pants and shirts: if I want 1 additional pair of pants I have to spend \$20 with which I could have bought **2** shirts.

Budget Constraint and Choice Set

[Graph]

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$I = \$200$, $P_s = \$10$, $P_p = \$20$

In general, the slope of a budget line is equal to “minus the price of the good on the horizontal axis divided by the price of the good on the vertical axis”.

Economic meaning: the slope of the budget line represents the **opportunity cost** of the good on the horizontal axis in terms of the good on the vertical axis (in our example how many shirts I have to give up for any additional pair of pants).

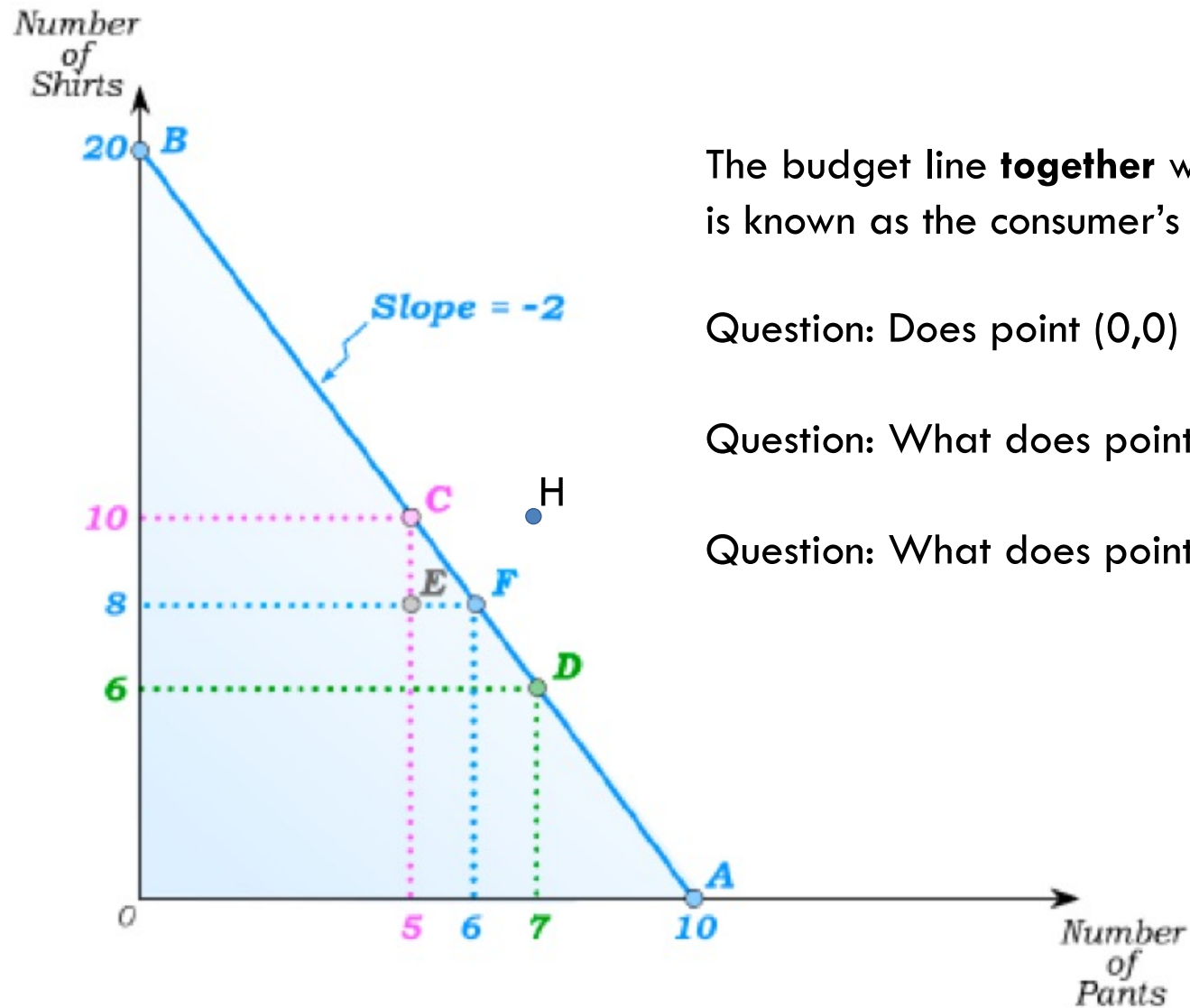
Opportunity costs are determined by the market prices of the two goods!

Question: What is the opportunity cost of shirts in terms of pants?

Budget Constraint and Choice Set

[Graph]

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The budget line **together** with the shaded area beneath it is known as the consumer's **choice set**.

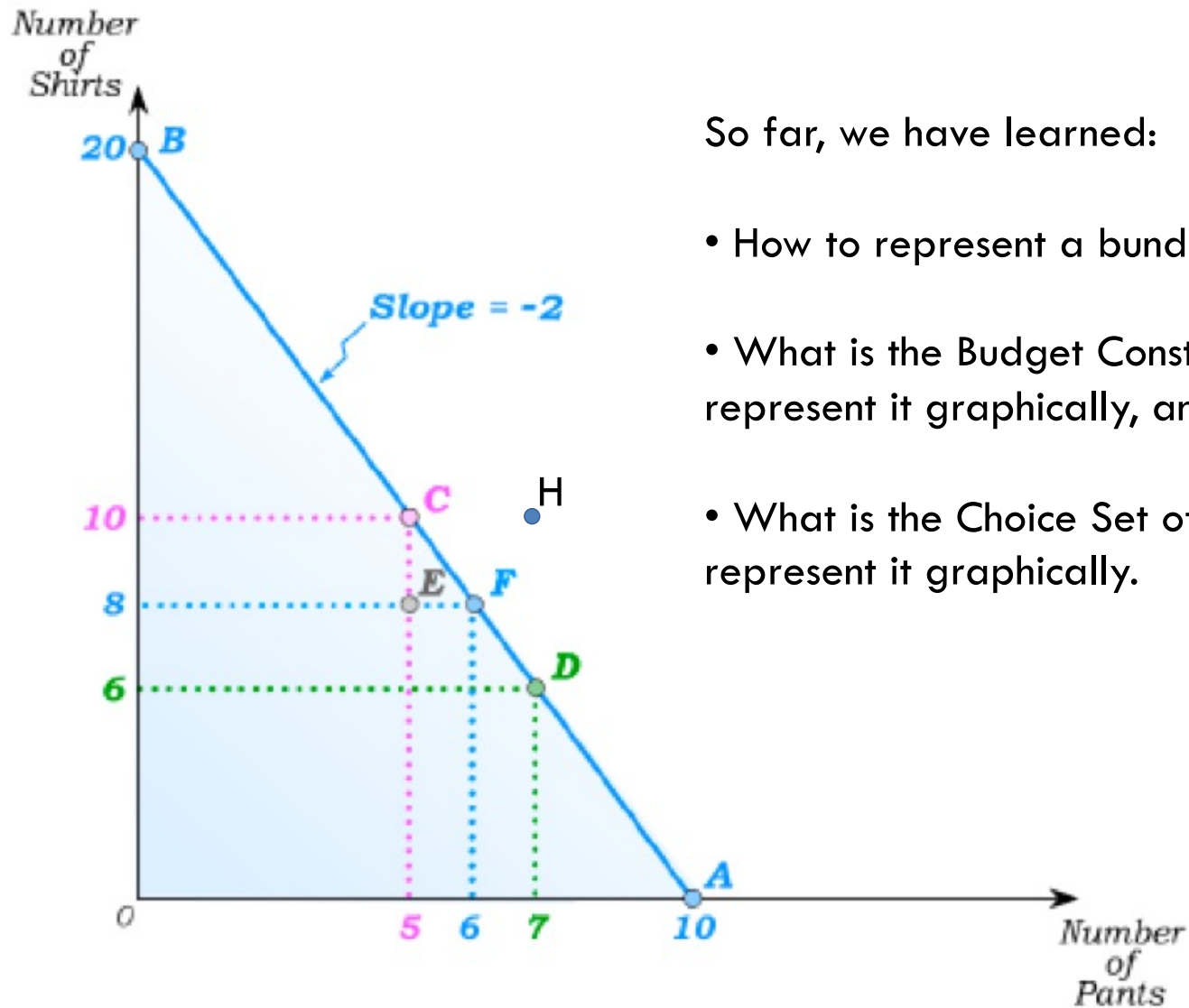
Question: Does point (0,0) belong to the choice set?

Question: What does point H represent?

Question: What does point E represent?

What have we learned so far?

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So far, we have learned:

- How to represent a bundle of two goods graphically.
- What is the Budget Constraint of a consumer, how to represent it graphically, and what its slope means.
- What is the Choice Set of a consumer and how to represent it graphically.

Budget Constraint and Choice Set

[Math]

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Let's stay in our world with only two goods: good 1 (pants) and good 2 (shirts).

Let x_1 represent the **quantity of good 1** (number of pants) and x_2 the **quantity of good 2** (number of shirts).

So (x_1, x_2) represents a bundle of x_1 pants and x_2 shirts that I may be interested in buying.

For example, If $x_1=5$ and $x_2=6$, I am talking about bundle $(5,6)$, i.e. a bundle of 5 pants and 6 shirts.

Budget Constraint and Choice Set

[Math]

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The problem we want to model mathematically is to find all the bundles of pants and shirts that I can afford to buy with my income.

Let p_1 represent the price of good 1 (i.e. the price of a pair of pants) and p_2 the price of good 2 (i.e. the price of a shirt).

Let I denote my income.

Budget Constraint and Choice Set

[Math]

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What is the **mathematical expression** representing the set of bundles that I can buy given my income I and prices p_1 and p_2 ?

To answer this question, let's first find out how we can write mathematically my **total** spending on a generic bundle (x_1, x_2) :

$$p_1x_1 + p_2x_2$$

Hence, the mathematical condition that represents all the bundles that I can buy with my income can be written as:

$$p_1x_1 + p_2x_2 \leq I$$

All bundles (x_1, x_2) that satisfy the previous inequality are in my choice set (i.e. I can buy them!)

Budget Constraint and Choice Set

[Math]

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Question: suppose $p_1 = \$20$, $p_2 = \$10$ and $I = \$200$.

Does bundle (3,4) belong to the choice set?

And bundle (5,10)?

Budget Constraint and Choice Set

[Math]

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We have found the mathematical expression of the choice set. What is the mathematical expression of the budget constraint (or budget line)?

Remember the definition of *Budget line* or *budget constraint*: The set of bundles that is affordable and that leaves me with no unspent income.

In our world with two goods, the mathematical expression of the “*budget line*” or “*budget constraint*” reads:

$$p_1x_1 + p_2x_2 = I$$

All bundles (x_1, x_2) that satisfy the previous equation are **on** the budget line (i.e., I can buy them and if I do so I will spend all my income!).

Math and Graph together

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Let's now make sense of these math expressions by showing that they are just representing what we have previously shown graphically!

We just saw that the math expression of the “**budget line**” or “**budget constraint**” reads:

$$p_1x_1 + p_2x_2 = I$$

Suppose I want to represent this budget constraint graphically.

First, I have to decide which good to put on the vertical axis and which on the horizontal.

Let's put good 1 (pants) on the horizontal and good 2 (shirts) on the vertical axis.

Then I re-write the budget constraint as follows:

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

Math and Graph together

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Number of Shirts (x_2)

$$\frac{I}{p_2}$$

$$\text{Slope} = -\frac{p_1}{p_2}$$

Horizontal intercept

$$\frac{I}{p_1}$$

Number of Pants (x_1)

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

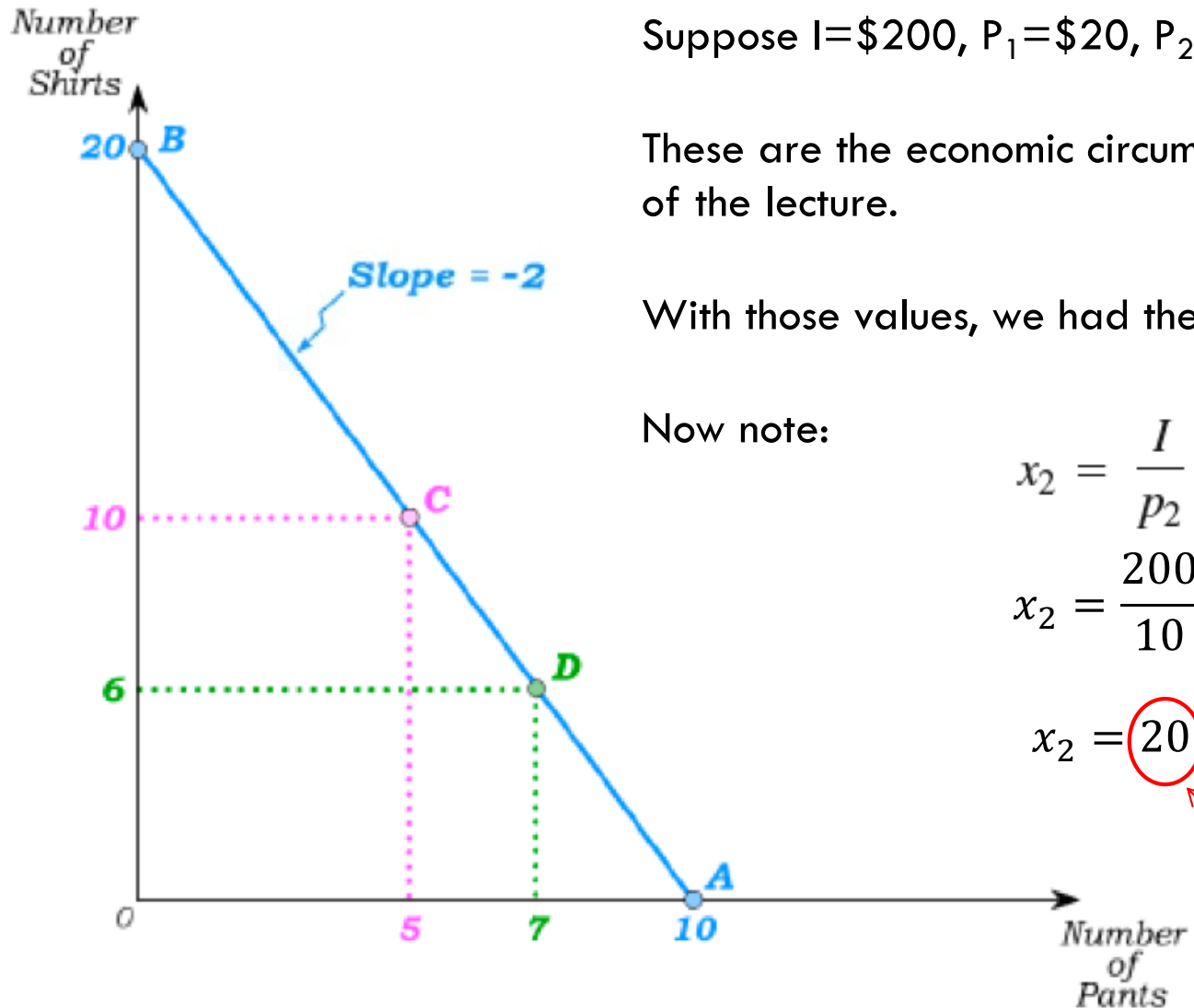
Slope

Vertical intercept

Remember: the slope of the budget line represents the *opportunity cost* of the good on the horizontal axis in terms of the good on the vertical axis.

Math and Graph together

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Suppose $I = \$200$, $P_1 = \$20$, $P_2 = \$10$.

These are the economic circumstances we used in the first part of the lecture.

With those values, we had the graph on the left (see slide 22).

Now note:

$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

$$x_2 = \frac{200}{10} - \frac{20}{10} x_1$$

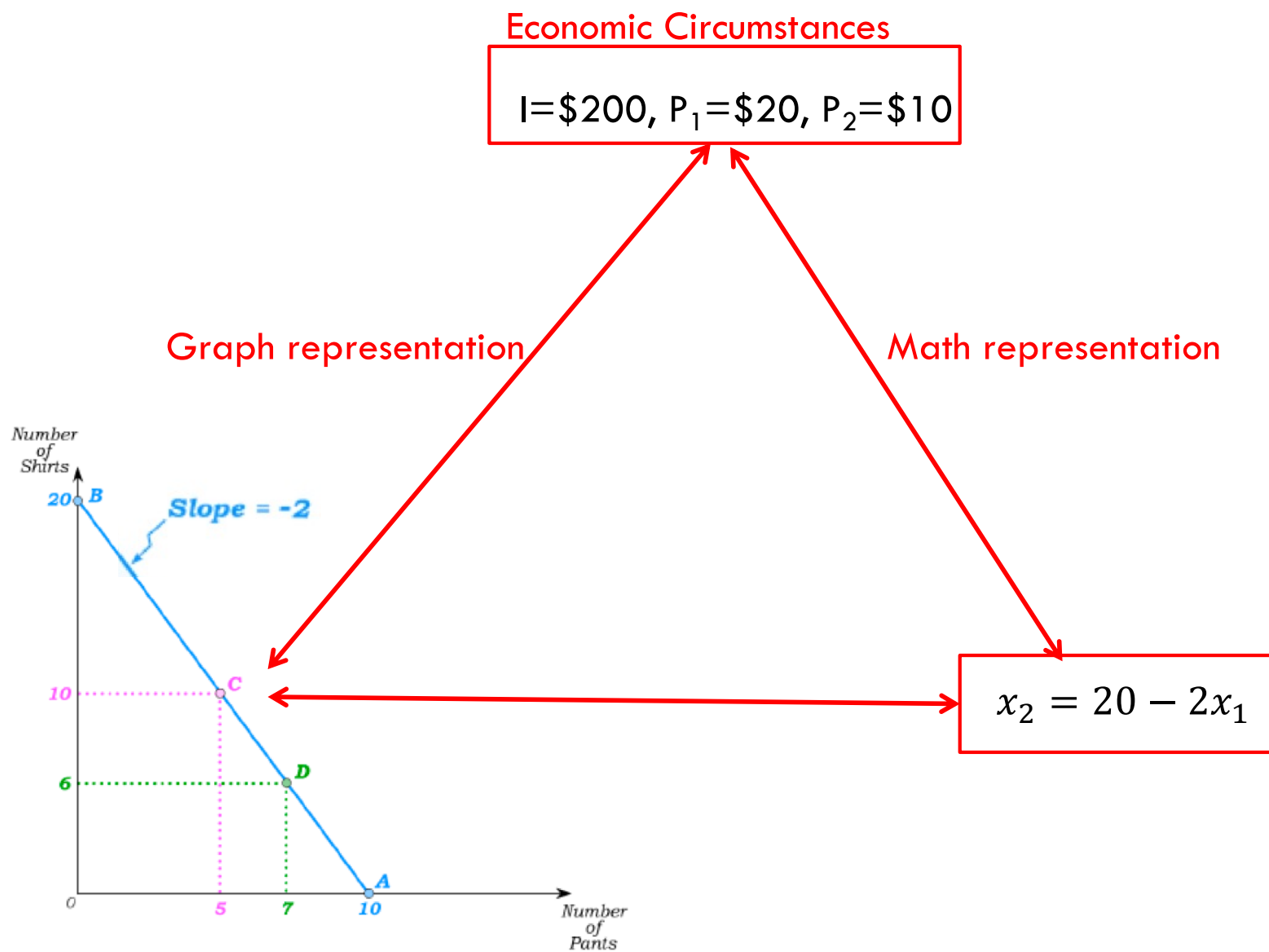
$$x_2 = 20 - 2x_1$$

Slope

Vertical intercept

Math and Graph together

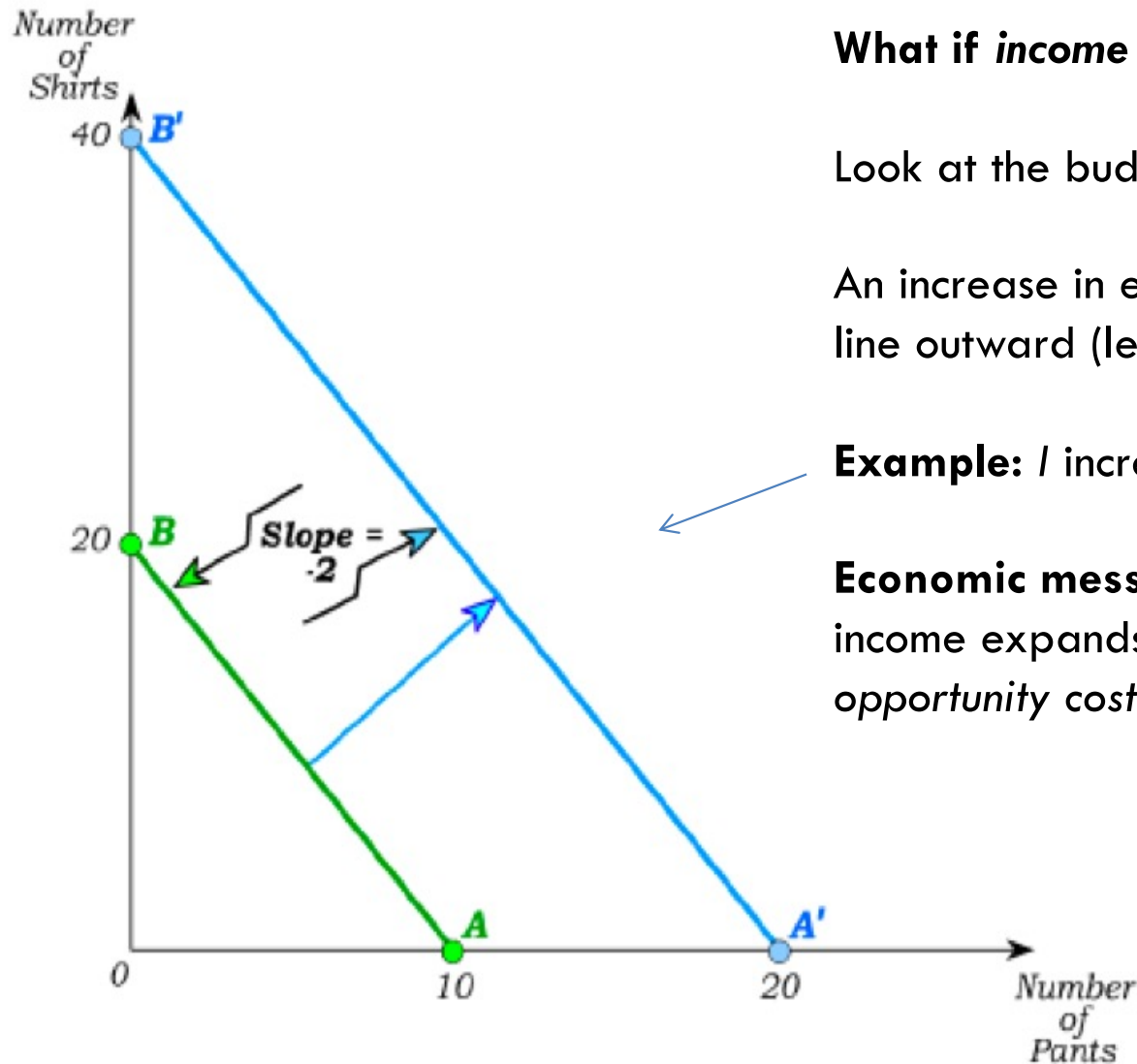
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A change in exogenous income

[Graph&Math]

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What if *income increases*? What changes?

Look at the budget line equation: $x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$

An increase in exogenous income shifts the budget line outward (leaving the slope unchanged)

Example: I increases from \$200 to \$400

Economic message: An increase in exogenous income expands the Choice Set without changing the opportunity costs.

A change in prices

[Graph&Math]

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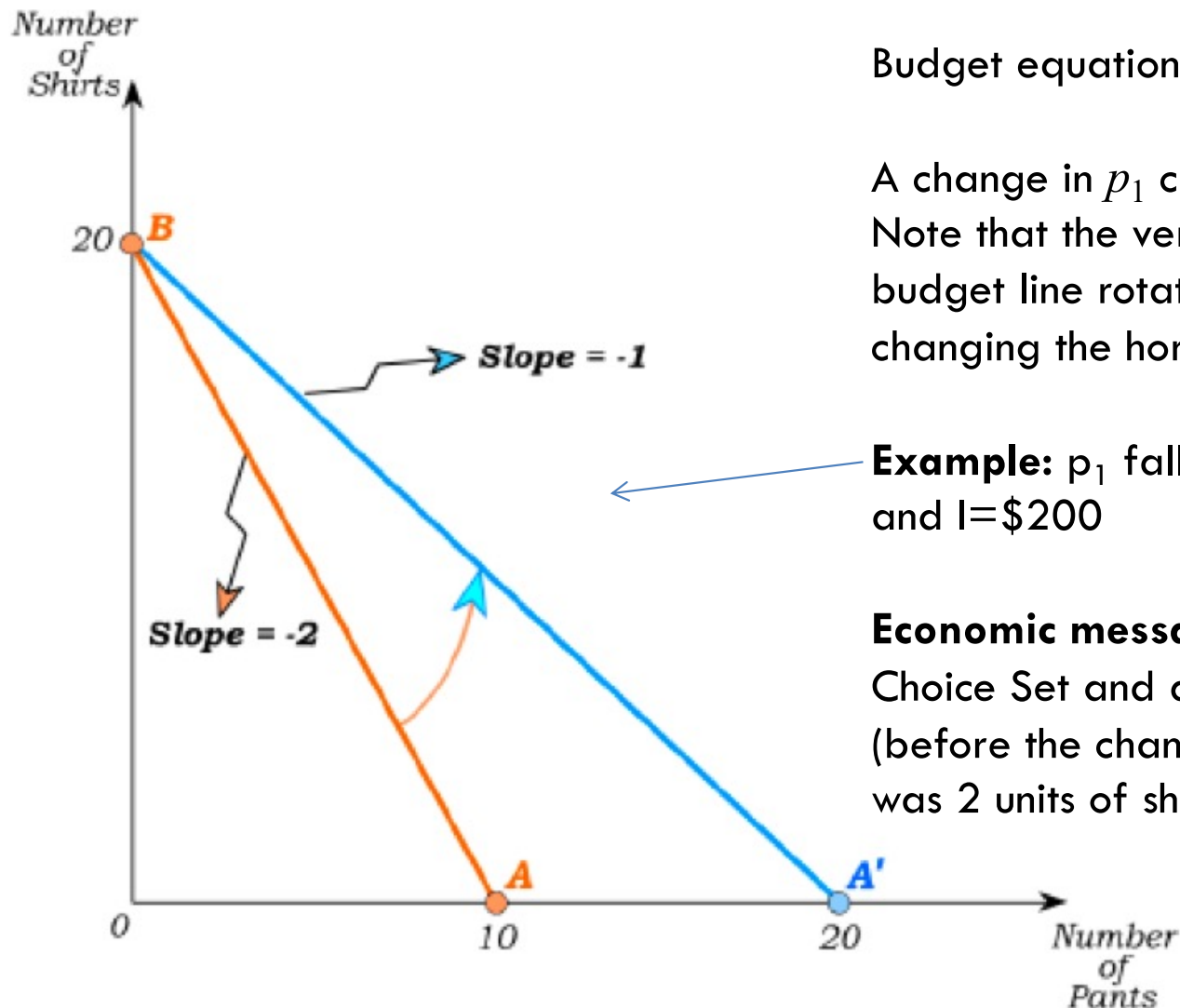
What if the price of pants (p_1) changes?

$$\text{Budget equation: } x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

A change in p_1 changes the slope of the budget line:
Note that the vertical intercept does not change: the budget line rotates around its vertical intercept (thus changing the horizontal intercept).

Example: p_1 falls from \$20 to \$10 (with $p_2 = \$10$ and $I = \$200$)

Economic message: A fall in p_1 expands the Choice Set and also changes the opportunity costs (before the change, the opportunity cost of pants was 2 units of shirts, after the change it is only 1 unit of shirts).



A change in prices

[Graph&Math]

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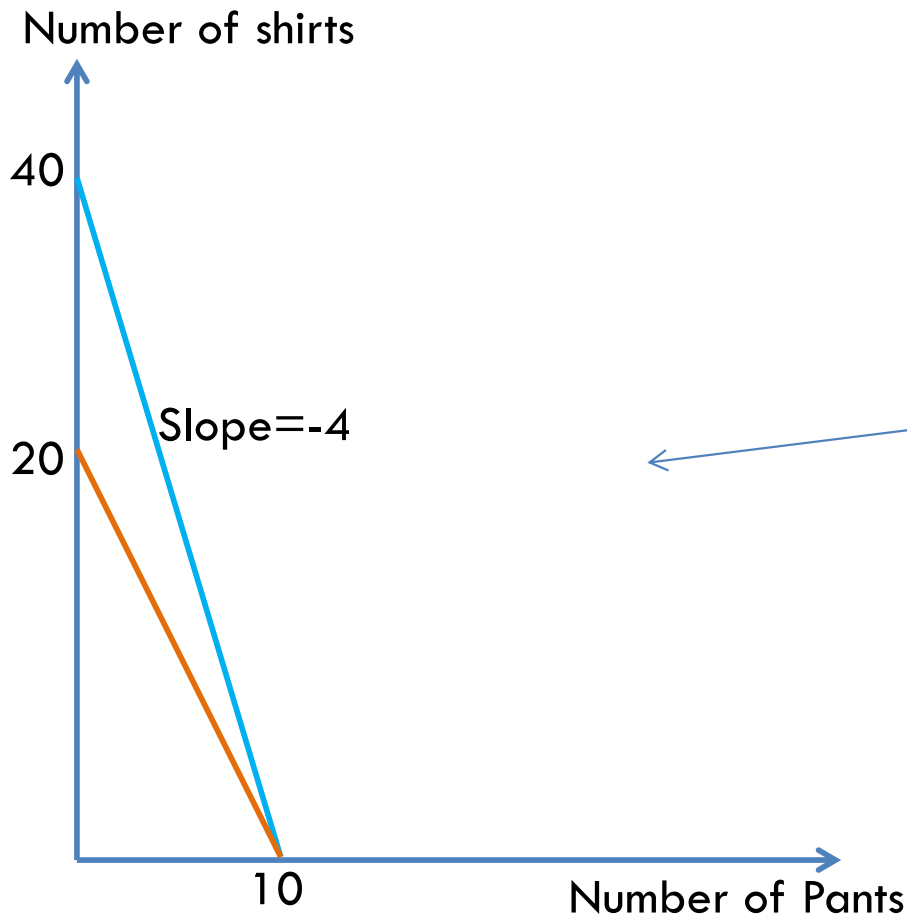
What if the price of shirts (p_2) changes?

Budget equation:
$$x_2 = \frac{I}{p_2} - \frac{p_1}{p_2} x_1$$

Slope changes and also vertical intercept. Indeed the budget line rotates around its horizontal intercept $\frac{I}{p_1}$

Example: p_2 falls from \$10 to \$5 (with $p_1 = \$20$ and $I = \$200$).

Economic message is the same as before! A fall in p_2 expands the Choice Set and also changes the opportunity costs (before the change, the opportunity cost of 1 unit of pants was 2 units of shirts, after the change it is 4 units of shirts)



A change in prices

[Conclusion]

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☐ **General conclusion:** A change in the price of one good (with all the rest unchanged) changes the Choice set by changing the opportunity costs of the two goods.

☐ **Questions:**

☐ What if prices of both goods change in the same proportion (e.g. both double)?

☐ What if prices and income change in the same proportion (for example, all double)?

Extensions

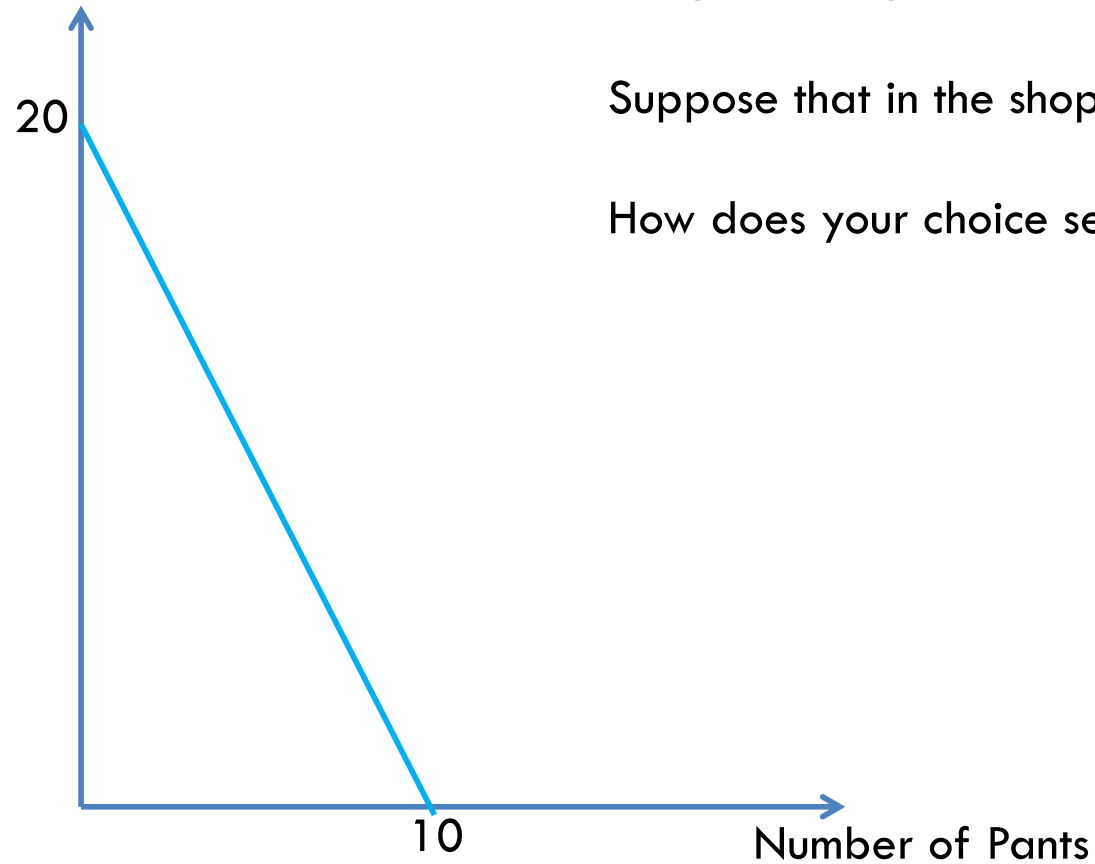
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- ☐ **Budget constraint with a quota**
- ☐ **Kinky Budget constraints**

Choice Sets with a Quota

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Number of shirts



The picture represents our original budget constraint.

Suppose that in the shop there are only 6 pants available.

How does your choice set change?

Choice Sets with a Quota

32

Number of shirts

20

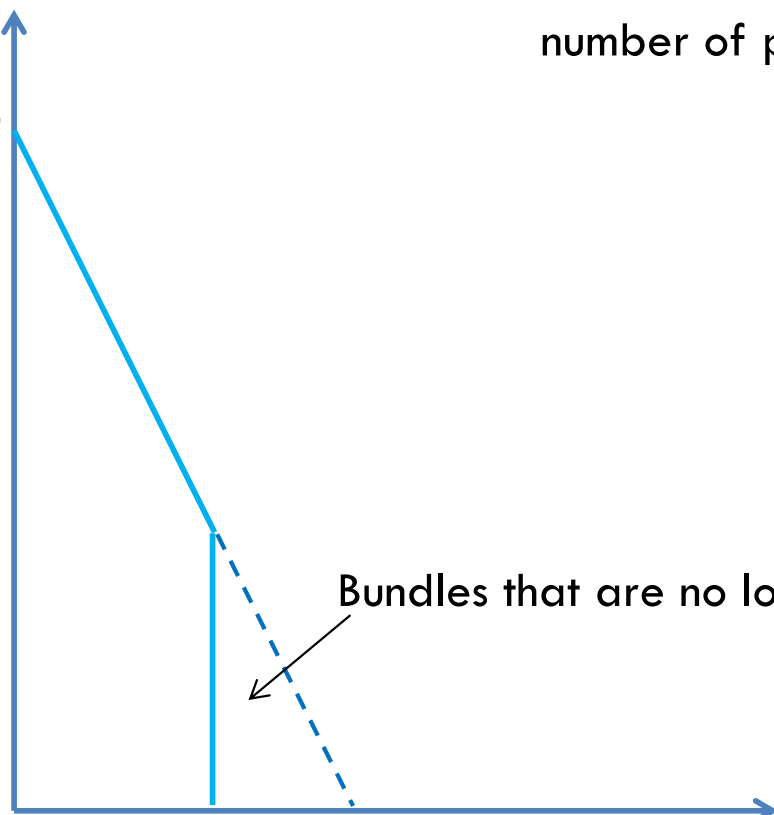
If there are only 6 pants available, the maximum number of pants that can be bought is 6.

6

10

Number of Pants

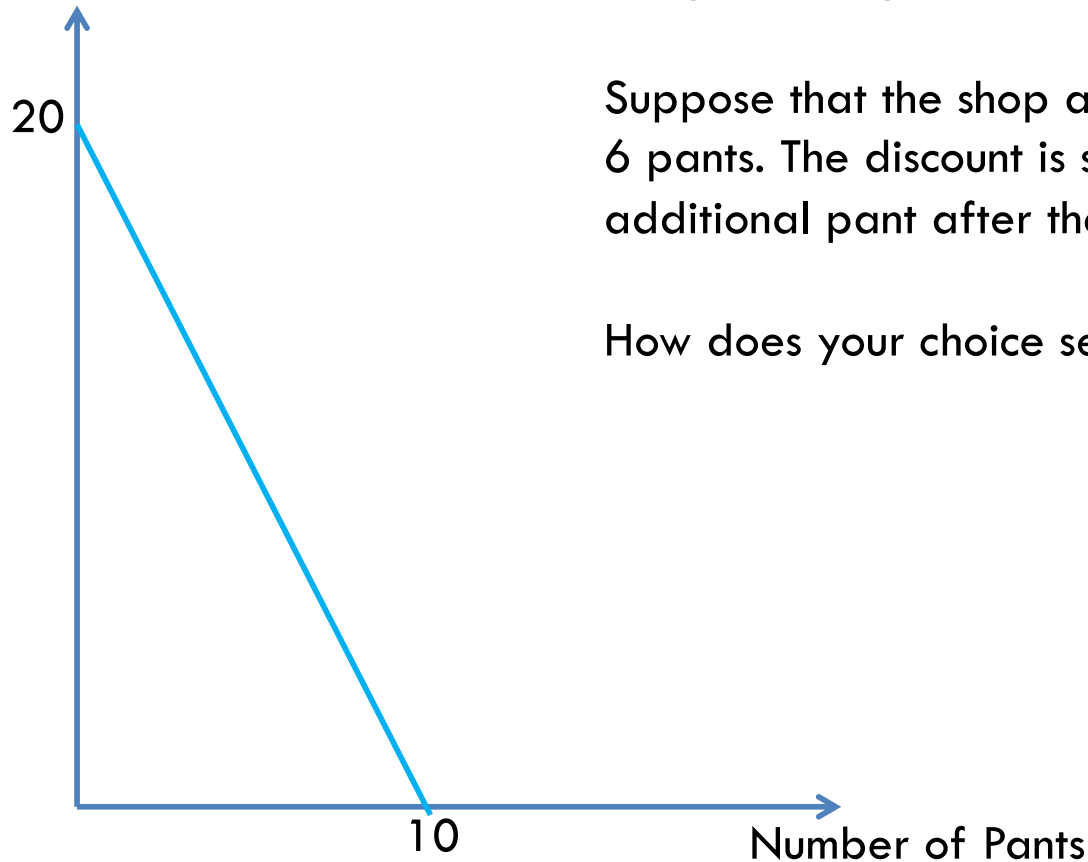
Bundles that are no longer available due to quota on pants



Choice Sets with Kinky Budget Constraints

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Number of shirts



The picture represents our original budget constraint.

Suppose that the shop applies a discount if you buy at least 6 pants. The discount is such that you get 50% off each additional pant after the initial 6 that you bought.

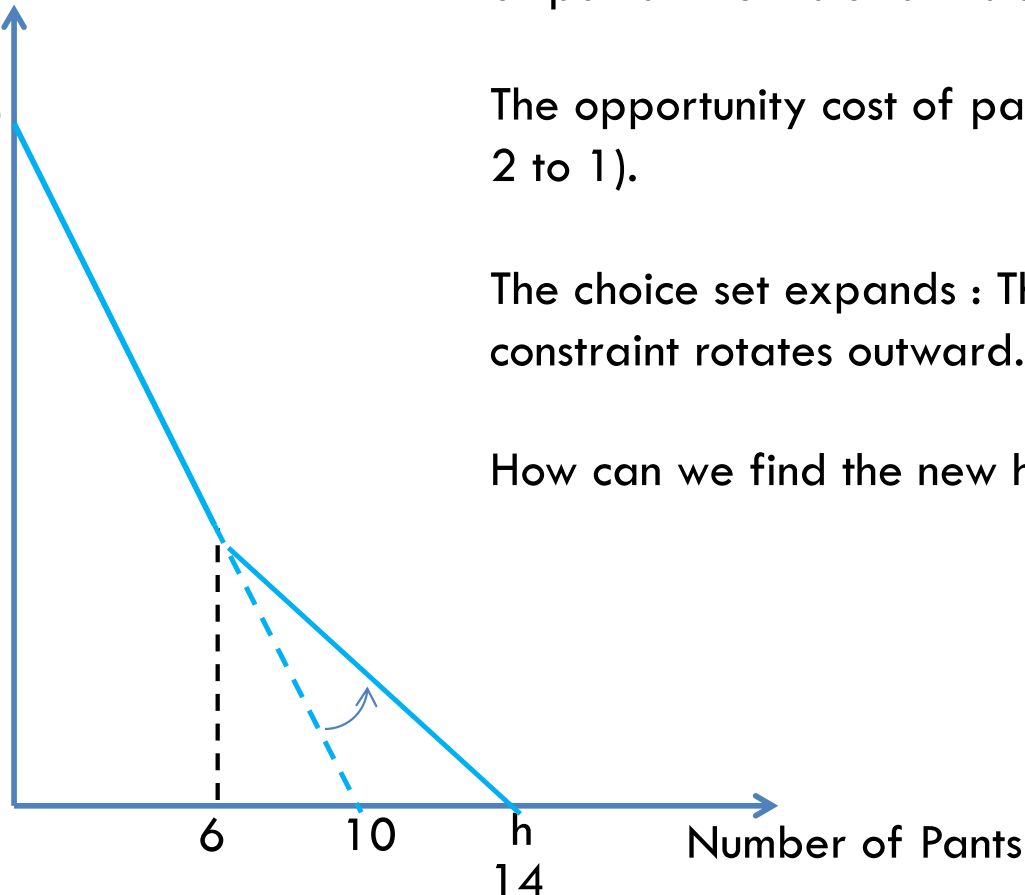
How does your choice set change?

Choice Sets with Kinky Budget Constraints

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Number of shirts

20



For each additional unit after the 6th one, the opportunity cost of pants in terms of shirts changes.

The opportunity cost of pants in terms of shirts decreases (from 2 to 1).

The choice set expands : The dashed-portion of the budget constraint rotates outward.

How can we find the new horizontal intercept?

$$6 \cdot 20 + (h - 6)10 = 200$$

$$h = 14$$

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End of Lecture